

The USGS Pacific Coastal and Marine Science Center (PCMSC) acquired a GeoTek Rotating X-Ray Computed Tomography (RXCT) scanner in 2018 for approximately \$650,000 and GeoTek Multi-Sensor Core Logger (MSCL) in 2003 (upgraded in 2012, and 2024) for approximately \$350,000 (amount includes upgrade costs). The RXCT is only CT machine across all USGS labs. By having both the RXCT and MSCL together in the same lab, the PCMSC Core Lab stands out as a top core lab in the United States. These machines routinely support research regarding, geologic hazard and mineral resource assessments across several USGS mission areas. Therefore, these GeoTek scanners not only help the USGS protect human life and infrastructure but also facilitate management of mineral resources. The required items and the technical specifications are the following:

- Thermo Kevex 130 Kv X-ray Source for RXCT S/N21
- Cs-137 Source for MSCL-S and MSCL-SXZ
- Disposal of Old Cs-137 Source
- Onsite installation/system realignment and training

1. The Thermo Kevex PXS10-WB X-ray source is the only source that Geotek uses in their Rotating X-ray Computed Tomography System.

2. Maximum tube voltage: 130 Kv

3. Operating volt range: 45-130 Kv

4. Tube current operational range: 0-500uA μ A

5. Maximum Power output: 65W

6. Minimum spot size of 6 microns at 4 watts, for high-resolution imaging

7. X-ray beam angle: 115°, round beam

8. Focus to object distance: 10.0 $\pm$ 0.5mm

9. Target material: Tungsten

10. X-ray output window material: Beryllium

11. Input power: 24-26 VDC, $\wedge$ A max

12. Control interface: RS-232

13. Small, round spot optimized over the range of operating voltage and power, for sharp, distortion-free images

14. 14mm spot to window spacing, for high geometric magnification

15. X-ray tube, power supply, and control electronics combined into one compact package, for easy system integration easy

16. Option to rotate the tube head +90° or -90° relative to home position, for flexible configuration and installation

17. Auto-conditioning, for protection of the source during start-up

18. Digital interface, for easy access to diagnostics and operating logs